Applying Probability and Statistics to Evaluate a Machine Learning Model

Alternative model: Random Forest

Objective. Determine whether an observed performance difference between a baseline classifier and an alternative classifier is large enough to be distinguished from plausible test-sample variation.

1. Problem Statement and Dataset

The dataset is the Breast Cancer Wisconsin Diagnostic dataset distributed with scikit-learn. It contains 569 observations and 30 numeric features computed from digitized images of breast-mass fine-needle aspirates. The target distinguishes malignant and benign cases. The data are loaded directly with `sklearn.datasets.load_breast_cancer`, so no manual download is required.

Baseline model: DecisionTreeClassifier with random_state=42. **Alternative model:** RandomForestClassifier with 300 trees, random_state=42, and n_jobs=1. Random Forest combines many decision trees trained with randomized samples/features; its superiority is tested rather than assumed.

2. Sampling and Experimental Methodology

A stratified 75%/25% train-test split is created with random_state=42. Stratification keeps class proportions approximately consistent across partitions. Both models are fitted only on the training set and evaluated on exactly the same held-out test rows (143 observations). The test set is not used to fit either model.

2.1 Sampling and estimation

Test accuracy is the sample proportion of correctly classified observations. It estimates classification accuracy on the population represented by the test sample. Because the estimate is based on a finite sample, it is subject to sampling variability.

2.2 Paired bootstrap confidence interval

For each test observation, the paired difference is +1 if Random Forest is correct and the baseline is wrong, −1 if the baseline is correct and Random Forest is wrong, and 0 if both models are either correct or wrong. The mean of these differences equals the observed accuracy difference. The code resamples test-row indices with replacement 10,000 times, applying the same sampled indices to both models, and uses the 2.5th and 97.5th percentiles for a 95% percentile bootstrap confidence interval.

2.3 Hypothesis test

The exact two-sided McNemar test is used because the models predict the same examples. *b* counts cases where the baseline is correct and Random Forest is wrong; *c* counts cases where the baseline is wrong and Random Forest is correct. Under the null hypothesis of equal error probabilities on discordant pairs, the smaller count is tested using an exact Binomial(n=b+c, p=0.5) distribution. The significance level is $\alpha = 0.05$.

H■: the classifiers have equal error probability on paired test cases. **H■:** their error probabilities differ. The test is two-sided because the question is whether performance differs, not only whether it changes in a preselected direction.

3. Observations, Decision, and Limitations

Measure	Baseline: Decision Tree	Alternative: Random Forest
Test accuracy	92.31%	95.80%
Correct predictions (out of 143)	132	137

Observed accuracy difference (Random Forest − Decision Tree): **+3.50 percentage points**. There are 2 discordant cases where the baseline is correct and Random Forest is wrong (b=2), and 7 cases where the baseline is wrong and Random Forest is correct (c=7).

The paired 95% bootstrap confidence interval is approximately −**0.70 to +7.69 percentage points**, which includes zero. The exact two-sided McNemar p-value is **0.1797**, greater than α=0.05.

Decision: Random Forest has higher observed test accuracy in this experiment, but the difference is not statistically significant at the 5% level. We fail to reject the null hypothesis; this does not prove the models are identical. The test sample does not provide strong enough evidence that the observed improvement is reliable rather than due to sampling variation.

Limitations

- The test set contains only 143 examples, so uncertainty remains and small differences are difficult to establish.
- A single train-test split can produce split-specific results. Repeated stratified cross-validation or repeated holdout evaluation could provide a more complete picture, while keeping a final test set untouched.
- The percentile bootstrap interval resamples observed test cases and cannot represent every source of uncertainty, such as changes in the future population or data collection process.
- Accuracy treats false positives and false negatives equally. In medical screening, sensitivity, specificity, calibration, and error costs should also be considered.
- Repeatedly tuning models against the same test set would cause test-set leakage and could make the reported result optimistic.

4. Where Probability and Statistics Are Used

- **Stratified random sampling:** split train/test data while preserving approximate class proportions.
- **Statistical estimation:** test accuracy estimates generalization performance from a sample.
- **Bootstrap resampling:** sampling test indices with replacement approximates the sampling distribution of the paired accuracy difference.
- **Confidence interval:** the 95% interval shows plausible values under the bootstrap procedure; because it includes zero, no improvement is compatible with the interval.
- **Hypothesis testing:** exact McNemar testing evaluates whether paired wins and losses are inconsistent with equal error probabilities.
- **p-value and significance level:** $p \approx 0.1797$ is compared with $\alpha = 0.05$; because $p > \alpha$, the null hypothesis is not rejected.
- **Probability model:** McNemar's exact test uses a Binomial(n=b+c, p=0.5) distribution under the null for discordant pairs.

Reproducibility

Run the Python script with Python 3 and numpy, scipy, and scikit-learn installed. The random seed, model settings, split, bootstrap replicate count, and statistical test are specified in the script.

```
pip install numpy scipy scikit-learn
python evaluate_ml_models.py
```

References

- scikit-learn documentation, load_breast_cancer dataset: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html
- scikit-learn documentation, RandomForestClassifier: https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html
- Fagerland, M. W., Lydersen, S., & Laake, K. (2013). The McNemar test for binary matched-pairs data: mid-p and exact tests. Statistical Methods in Medical Research, 22(4), 346–361.